Fiscal Year:	FY 2004	**Task Last Updated:**	FY 10/13/2004
PI Name:	Urban, David L. Ph.D.		
Project Title:	Characterization of Smoke from Microgravity Fires for Improved Spacecraft Fire Detection		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	NASA Glenn Research Center		
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City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	10
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	03/02/2001	**End Date:**	02/28/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mulholland, George (National Institute for Standards and Technology) Yang, Jiann (National Institute for Standards and Technology) Cleary, Thomas (National Institute for Standards and Technology) Yuan, Zeng guang (National Center for Microgravity Research)		
Grant/Contract No.:	IH1013215		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The project is a flight-definition investigation funded under the Bio-Astronautics initiative. The team is a combination of NASA Glenn Research personnel with experience in microgravity combustion and researchers from the National Institute for Standards and Technology who have extensive experience in smoke detection. The investigation will define an experiment that will make measurements of the particle size distribution of spacecraft smoke from a variety of spacecraft materials. The overall project objective is to improve the reliability of spacecraft fire detection by making measuremstns of the particle size distribution of spacecraft smoke. The smoke detectors currently used in NASA spacecraft are based upon understanding from 1-g smoke measurements. Experiments conducted aboard the space shuttle in 1996 demonstrated that the particle size distribution for various materials can change substantially and that smoke detector sensitivity in 1-g is not necessarily a good indicator for performance in low-gravity. This suggests that the reliability of spacecraft smoke detectors can be substantially improve dif they are designed based upon the smoke particulate properties of spacecraft smoke.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Science concept definition: The investigation is in the science concept definition phase. A final draft of the Science Requirements Document has been written and subjected to internal review. At this point the project is ready for peer review once the panel is selected by Headquarters. Development of flight diagnostics: One of the critical diagnostics for the Smoke experiment is a particle number counting device. Numerous systems exist but are limited by potential gravitational susceptibility. Working with the project team, a modified P-Trak- a type of CNC (condensation nuclei counter) has been developed which is believed will operate effectively in low-gravity. Specifically, the condenser in the unit has been modified to induce the flow of liquid alcohol back towards the saturator, largely independent of gravitational forces. This is accomplished through a series of precisely machined grooves which encourage the wicking of the liquid alcohol in the proper direction along the condenser's inner wall. The design was developed following an analysis to measure the contact angle of isopropyl alcohol on the aluminum surfaces of the condenser. The hardware has been prepared for flight as a risk mitigation experiment and is ready for launch when upmass becomes available. Aerosol Dynamics modeling: Two aerosol growth dynamics model have been developed at NIST. One model is a zero-dimensional model for a nucleating/condensing vapor that is isothermal and spatially uniform and the other model employs the NIST Fire Dynamics Simulator code. Both models have been used to predict smoke growth in the experiment hardware and have produced mutually consistent predictions of the condensation rate. Corresponding validation experiments are underway.
Bibliography Type:	Description: (Last Updated: 02/27/2004)